

DBMS Lab — Week 1: Building a Foundation-Level Taxation Database

Learning Objective

By the end of this lab, students should be able to:

- Create a database using `CREATE DATABASE`.
 - Select a database using `USE`.
 - Create tables using `CREATE TABLE`.
 - Choose appropriate SQL data types.
 - Apply `PRIMARY KEY`, `NOT NULL`, and `UNIQUE` constraints.
 - Insert multiple records using `INSERT`.
 - Modify existing records using `UPDATE`.
 - Remove records using `DELETE`.
 - Modify table structure using `ALTER TABLE`.
 - Understand how real-world taxation information can be represented as database tables.
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1. Scenario

Imagine you are part of a software development team building a **basic taxation information system for India**.

The complete taxation system may eventually need to manage:

- taxpayers,
- different sources of income,
- financial years,
- tax returns,
- deductions,
- tax payments,
- and other information.

However, the system will be built **incrementally throughout the DBMS lab**.

For **Week 1**, your task is to create the basic database and store foundational data using four independent entities:

1. Taxpayer
2. Income_Category
3. Financial_Year
4. Income_Record

Important: Relationships and foreign keys are intentionally not included in Week 1. They will be introduced in a later lab.

Entity 1: Taxpayer

A **taxpayer** is an individual whose personal and identification information is maintained by the taxation system.

Create a table named:

Taxpayer

Required attributes

Column	Suggested Data Type	Requirement
taxpayer_id	INT	Primary Key
pan_number	VARCHAR(10)	Not Null, Unique
full_name	VARCHAR(100)	Not Null
date_of_birth	DATE	Not Null
occupation	VARCHAR(50)	Not Null

annual_income	DECIMAL(12,2)	Not Null
email	VARCHAR(100)	Unique
is_active	BOOLEAN	—

Data to be inserted

taxpayer_id	pan_number	full_name	date_of_birth	occupation	annual_income	email	is_active
101	ABCDE1234F	Ravi Kumar	1995-06-15	Software Engineer	850000.00	ravi.kumar@example.com	TRUE
102	BCDEF2345G	Priya Sharma	1992-11-22	Doctor	120000.00	priya.sharma@example.com	TRUE
103	CDEFG3456H	Arjun Reddy	1988-03-10	Business Owner	180000.00	arjun.reddy@example.com	TRUE
104	DEFGH4567J	Sneha Patel	1998-08-05	Teacher	620000.00	sneha.patel@example.com	TRUE

						xample.com	
105	EFGHJ 5678K	Kiran Rao	1990-0 1-18	Freelan cer	750000. 00	kiran.ra o@exa mple.co m	TRUE
106	FGHJK 6789L	Meera Singh	1985-1 2-30	Consult ant	150000 0.00	meera.s ingh@e xample. com	FALSE

What students learn from this table

This single table introduces several data types:

INT
VARCHAR
DATE
DECIMAL
BOOLEAN

and constraints:

PRIMARY KEY
NOT NULL
UNIQUE

Entity 2: Income Category

People can earn income from different sources. A taxation database needs to classify these income sources.

Create a table named:

Income_Category

Required attributes

Column	Suggested Data Type	Requirement
category_id	INT	Primary Key
category_name	VARCHAR(50)	Not Null, Unique
description	VARCHAR(200)	Not Null
taxable	BOOLEAN	Not Null

Data to be inserted

category_id	category_name	description	taxable
1	Salary	Income received from employment	TRUE
2	Business	Income earned from business activities	TRUE
3	House Property	Income received from property or rent	TRUE

4	Capital Gains	Income from transfer of eligible assets	TRUE
5	Other Sources	Income such as bank interest	TRUE
6	Agricultural Income	Income from eligible agricultural activities	FALSE

Learning purpose

Students understand that databases often maintain **category/master tables** instead of repeatedly entering descriptive information.

At this stage, however, students do **not** create a foreign key relationship.

Entity 3: Financial Year

Tax-related information is organized by financial year.

Create a table named:

Financial_Year

Required attributes

Column	Suggested Data Type	Requirement
year_id	INT	Primary Key

year_label	VARCHAR(9)	Not Null, Unique
start_date	DATE	Not Null
end_date	DATE	Not Null
filing_deadline	DATE	—
is_current	BOOLEAN	Not Null

Data to be inserted

year_id	year_label	start_date	end_date	filing_dead line	is_current
1	2020-2021	2020-04-01	2021-03-31	2021-07-31	FALSE
2	2021-2022	2021-04-01	2022-03-31	2022-07-31	FALSE
3	2022-2023	2022-04-01	2023-03-31	2023-07-31	FALSE
4	2023-2024	2023-04-01	2024-03-31	2024-07-31	FALSE
5	2024-2025	2024-04-01	2025-03-31	2025-07-31	FALSE

6	2025-2026	2025-04-01	2026-03-31	2026-07-31	TRUE
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This reinforces:

INT
VARCHAR
DATE
BOOLEAN

Entity 4: Income Record

This table represents individual income records.

For now, students will store `taxpayer_id`, `category_name`, and `financial_year` as ordinary values. **They are not foreign keys yet.**

Later, students can redesign this table and establish proper relationships.

Create a table named:

Income_Record

Required attributes

Column	Suggested Data Type	Requirement
income_id	INT	Primary Key
taxpayer_id	INT	Not Null
income_source	VARCHAR(100)	Not Null

category_name	VARCHAR(50)	Not Null
amount	DECIMAL(12,2)	Not Null
received_date	DATE	Not Null
financial_year	VARCHAR(9)	Not Null

Data to be inserted

income_id	taxpayer_id	income_source	category_name	amount	received_date	financial_year
1001	101	TechNova Solutions	Salary	850000.00	2026-03-31	2025-2026
1002	102	City Care Hospital	Salary	1200000.00	2026-03-31	2025-2026
1003	103	Reddy Enterprises	Business	1800000.00	2026-03-31	2025-2026
1004	104	Sunrise School	Salary	620000.00	2026-03-31	2025-2026

1005	105	Web Design Projects	Business	750000.00	2026-03-31	2025-2026
1006	106	Professional Consulting	Business	1500000.00	2026-03-31	2025-2026

Week 1 Student Task

Part A — Database Creation

Create a database named:

taxation_db

Select the database for use.

Students should determine and write the required SQL statements themselves.

Part B — Table Creation using DDL

Create the following four tables:

Taxpayer
Income_Category
Financial_Year
Income_Record

Requirements:

- Select appropriate data types.
- Define one **PRIMARY KEY** for each table.
- Apply **NOT NULL** where values are mandatory.

- Apply **UNIQUE** to values that should not be duplicated.
 - Do not use **FOREIGN KEY** in Week 1.
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Part C — Data Insertion using DML

Insert all the provided records into the four tables.

Students may use either:

```
INSERT INTO table_name VALUES (...);
```

or:

```
INSERT INTO table_name (column1, column2, ...)  
VALUES (...);
```

They should preferably practice the second form because it explicitly identifies the columns being populated.

Part D — DML Tasks

After inserting the initial data, perform the following operations.

1. **INSERT:** Add a new taxpayer with **taxpayer_id = 107**. Choose appropriate values for all attributes.
2. **UPDATE:** Ravi Kumar's annual income has increased from **850000.00** to **950000.00**. Update the appropriate record.
3. **UPDATE:** Kiran Rao's occupation has changed from **Freelancer** to **Software Consultant**.
4. **UPDATE:** Activate the taxpayer account of Meera Singh by changing **is_active** from **FALSE** to **TRUE**.
5. **DELETE:** Delete the newly created taxpayer with **taxpayer_id = 107**.
6. **INSERT:** Add a new income category named **Rental Income** with an appropriate description.

Students should write the SQL statements themselves rather than being given the completed commands.

Part E — DDL Modification Tasks

After creating and populating the database, modify the table structures.

Task 1: Add a new column

Add the following column to the `Taxpayer` table:

`phone_number`

Choose an appropriate data type.

Task 2: Add another column

Add the following column to the `Income_Record` table:

`remarks`

Choose an appropriate data type.

Task 3: Modify a column

Modify the size of `occupation` in the `Taxpayer` table so that it can store longer occupation descriptions.

Task 4: Create a temporary table

Create a table named:

`Tax_Office`

with:

Column	Requirement
<code>office_id</code>	Primary Key

office_name	Not Null
city	Not Null

Insert two sample records.

Task 5: Practice

TRUNCATE

Remove **all records** from `Tax_Office` while retaining the table structure.

Task 6: Practice

DROP

Delete the `Tax_Office` table completely from the database.

This makes the distinction very clear:

DELETE → Removes selected rows

TRUNCATE → Removes all rows but keeps the table

DROP → Removes the entire table

Part F — Constraint Observation Tasks

These are particularly important. Students should intentionally perform operations that fail and observe the database error.

Experiment 1 — Duplicate Primary Key

Try inserting another taxpayer with:

taxpayer_id = 101

Question: What error occurs and why?

Experiment 2 — Duplicate Unique Value

Try inserting another taxpayer using:

```
pan_number = ABCDE1234F
```

Question: Why does the database reject the record?

Experiment 3 — NULL in a NOT NULL column

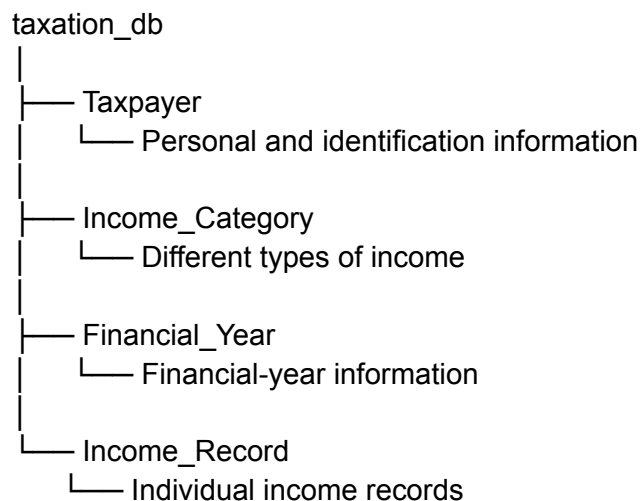
Try inserting a taxpayer without a `full_name`.

Question: What constraint prevents the insertion?

This helps students understand that constraints are not merely SQL syntax—they enforce **data integrity**.

Expected Week 1 Outcome

By the end of the lab, the student should have a database containing:



The important design decision for **Week 1** is that `Income_Record` contains values such as:

```
taxpayer_id = 101  
category_name = 'Salary'  
financial_year = '2025-2026'
```

but the database does **not yet enforce relationships** between these values and the other tables.

That creates a useful transition for a later week:

“We have created multiple tables, but how does the database know that taxpayer 101 actually exists? How do we prevent an income record from referring to taxpayer 999, who does not exist?”